Lesson 43: VI Error Bounds and Monotonicity

AI/ML Mathematics

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1 Motivation and Overview

Lesson 42 established that value iteration converges geometrically to V^* and provided basic a priori and a posteriori error bounds. In practice, however, we need sharper tools. Consider running value iteration on a large MDP with $\gamma = 0.99$: the a priori bound suggests nearly a thousand iterations are needed for moderate accuracy. Can we do better? Can we detect convergence earlier by exploiting the structure of the iterates?

This lesson develops refined error analysis for value iteration. We derive tighter two-sided bounds using the span seminorm, establish monotonicity properties of VI sequences under specific initializations, and show how these properties lead to improved stopping criteria. We also analyze the rate at which the greedy policy stabilizes – often the policy converges long before the value function does, suggesting that tracking policy changes provides a more efficient termination test.

Why Refined VI Analysis Matters for RL/ML

- **Practical stopping rules**: Naive stopping criteria based on the a posteriori bound can be overly conservative by a factor of 2; the span-based bounds developed here tighten this significantly.
- **Early policy convergence**: In many RL applications, the policy (greedy actions) stabilizes well before the value function converges to full precision, motivating policy-based stopping criteria.
- **Monotone initialization**: Initializing value iteration with known bounds (e.g., $v^{(0)} = 0$ or $v^{(0)} = R_{\max}/(1-\gamma)$) yields monotone sequences that bracket V^* from below and above, enabling interval estimates.
- **Approximate DP**: Error bounds for VI directly inform the analysis of fitted value iteration (FVI) used in deep RL, where each backup is approximate.

Prerequisites. Value iteration algorithm, a priori and a posteriori error bounds, approximate policy extraction, the Bellman optimality operator $\mathcal{T}^*$ and its contraction and monotonicity properties (Lesson 42). The Banach fixed-point theorem (Lesson 4, Lesson 15). Supremum norm on $\mathbb{R}^S$ (Lesson 5).

Learning objectives. After completing this lesson, the student will be able to:

- *Define* the span seminorm and *prove* that $\mathcal{T}^*$ contracts the span.
- *Derive* span-based error bounds for value iteration that improve upon the standard a posteriori bound.
- *Prove* monotonicity of value iteration sequences under appropriate initialization.
- *Construct* upper and lower bounding sequences that bracket V^*.
- *Analyze* the rate of policy convergence relative to value convergence.
- *Apply* refined stopping criteria to determine when value iteration can be terminated.

2 References

- Puterman [2014, Chapter 6, Sections 6.3–6.6] – Detailed treatment of span-based bounds, monotonicity, and stopping criteria for value iteration.

- Bertsekas [2012, Chapter 1] – Error bounds and approximate policy extraction in the context of approximate dynamic programming.
- Denardo [1967] – Contraction framework providing the theoretical basis for span contraction results.
- Sutton and Barto [2018] – Practical perspective on convergence detection in value iteration.

3 The Span Seminorm

The standard a posteriori bound from Lesson 42 uses the full supremum norm difference $\|v^{(n)} - v^{(n-1)}\|_\infty$. The span seminorm captures only the "spread" of a vector, ignoring uniform shifts, and leads to tighter bounds.

Definition 3.1 (Span Seminorm)**.** The *span seminorm* of a vector $v \in \mathbb{R}^S$ is

$$\operatorname{sp}(v) = \max_{s\in\mathcal{S}} v(s) - \min_{s\in\mathcal{S}} v(s).$$

Proposition 3.2 (Properties of the Span)**.** *For all $u, v \in \mathbb{R}^S$ and $c \in \mathbb{R}$:*

1. $\operatorname{sp}(v) \geq 0$*, with equality if and only if v is constant.*
2. $\operatorname{sp}(v + c\mathbf{1}) = \operatorname{sp}(v)$*, where $\mathbf{1} = (1, \ldots, 1)^\top$ (translation invariance).*
3. $\operatorname{sp}(\alpha v) = |\alpha| \operatorname{sp}(v)$ *for $\alpha \in \mathbb{R}$ (absolute homogeneity).*
4. $\operatorname{sp}(u + v) \leq \operatorname{sp}(u) + \operatorname{sp}(v)$ *(subadditivity).*
5. $\operatorname{sp}(v) \leq 2\|v\|_\infty$.
6. $\operatorname{sp}(u - v) \leq 2\|u - v\|_\infty$.

Proof. Properties (1)–(4) follow directly from the definition. For (5): $\max_s v(s) \leq \|v\|_\infty$ and $\min_s v(s) \geq -\|v\|_\infty$, so $\operatorname{sp}(v) \leq 2\|v\|_\infty$. Property (6) follows from (5) applied to $u - v$. □

Remark 3.3. The span is a *seminorm*, not a norm: $\operatorname{sp}(v) = 0$ does not imply $v = 0$, only that v is a constant vector. This is precisely the property that makes it useful: the Bellman operator shifts values uniformly by the reward, and the span ignores this shift.

Theorem 3.4 (Span Contraction of $\mathcal{T}^*$)**.** *The Bellman optimality operator satisfies*

$$\operatorname{sp}(\mathcal{T}^* u - \mathcal{T}^* v) \leq \gamma \operatorname{sp}(u - v)$$

for all $u, v \in \mathbb{R}^S$.

Proof. Let $w = u - v$. For any states $s_1, s_2 \in \mathcal{S}$:

$$\begin{aligned}(\mathcal{T}^* u)(s_1) - (\mathcal{T}^* v)(s_1) = \max_a &\left[R(s_1, a) + \gamma \sum_{s'} P(s' \mid s_1, a) u(s')\right] \\ &- \max_a \left[R(s_1, a) + \gamma \sum_{s'} P(s' \mid s_1, a) v(s')\right].\end{aligned}$$

Let $a_1^* = \arg\max_a [R(s_1, a) + \gamma \sum_{s'} P(s' \mid s_1, a) u(s')]$. Then

$$\begin{aligned}(\mathcal{T}^* u)(s_1) - (\mathcal{T}^* v)(s_1) &\leq R(s_1, a_1^*) + \gamma \sum_{s'} P(s' \mid s_1, a_1^*) u(s') \\ &\quad - R(s_1, a_1^*) - \gamma \sum_{s'} P(s' \mid s_1, a_1^*) v(s') \\ &= \gamma \sum_{s'} P(s' \mid s_1, a_1^*) w(s') \\ &\leq \gamma \max_{s'} w(s').\end{aligned}$$

Similarly, let $a_2^* = \arg\max_a [R(s_2, a) + \gamma \sum_{s'} P(s' \mid s_2, a) v(s')]$. Then

$$(\mathcal{T}^* u)(s_2) - (\mathcal{T}^* v)(s_2) \geq \gamma \sum_{s'} P(s' \mid s_2, a_2^*) w(s') \geq \gamma \min_{s'} w(s').$$

Therefore

$$\begin{aligned}[(\mathcal{T}^* u)(s_1) - (\mathcal{T}^* v)(s_1)] - [(\mathcal{T}^* u)(s_2) - (\mathcal{T}^* v)(s_2)] &\leq \gamma[\max_{s'} w(s') - \min_{s'} w(s')] \\ &= \gamma \operatorname{sp}(w).\end{aligned}$$

Taking the maximum over s_1 and minimum over s_2 of the left-hand side gives $\operatorname{sp}(\mathcal{T}^* u - \mathcal{T}^* v) \leq \gamma \operatorname{sp}(u - v)$. □

4 Span-Based Error Bounds

We now derive the improved error bounds using the span.

Theorem 4.1 (Span-Based A Posteriori Bound)**.** *Let $\{v^{(n)}\}$ be the value iteration sequence. Define the* Bellman residual *$h^{(n)} = v^{(n)} - v^{(n-1)}$ and let*

$$\underline{h}^{(n)} = \min_s h^{(n)}(s), \quad \overline{h}^{(n)} = \max_s h^{(n)}(s).$$

Then for all $n \geq 1$ and all $s \in \mathcal{S}$:

$$v^{(n)}(s) + \frac{\underline{h}^{(n)}}{1-\gamma} \leq V^*(s) \leq v^{(n)}(s) + \frac{\overline{h}^{(n)}}{1-\gamma}.$$

Consequently,

$$\|v^{(n)} - V^*\|_\infty \leq \frac{\operatorname{sp}(h^{(n)})}{2(1-\gamma)} + \frac{|\overline{h}^{(n)} + \underline{h}^{(n)}|}{2(1-\gamma)}.$$

Proof. **Upper bound.** Since $h^{(n)}(s) \leq \overline{h}^{(n)}$ for all s, we have $v^{(n)} \leq v^{(n-1)} + \overline{h}^{(n)}\mathbf{1}$. By monotonicity and the shift property of $\mathcal{T}^*$ (Lesson 42, Propositions 8.1 and 8.2):

$$\begin{aligned}v^{(n+1)} = \mathcal{T}^* v^{(n)} &\leq \mathcal{T}^*(v^{(n-1)} + \overline{h}^{(n)}\mathbf{1}) \\ &= \mathcal{T}^* v^{(n-1)} + \gamma\, \overline{h}^{(n)}\mathbf{1} \\ &= v^{(n)} + \gamma\, \overline{h}^{(n)}\mathbf{1}.\end{aligned}$$

Thus $h^{(n+1)}(s) = v^{(n+1)}(s) - v^{(n)}(s) \leq \gamma\, \overline{h}^{(n)}$.

Iterating: for $k \geq 1$, $\overline{h}^{(n+k)} \leq \gamma^k \overline{h}^{(n)}$. Therefore

$$\begin{aligned}V^*(s) - v^{(n)}(s) &= \sum_{k=0}^{\infty} [v^{(n+k+1)}(s) - v^{(n+k)}(s)] = \sum_{k=0}^{\infty} h^{(n+k+1)}(s) \\ &\leq \sum_{k=0}^{\infty} \overline{h}^{(n+k+1)} \leq \sum_{k=0}^{\infty} \gamma^{k+1} \overline{h}^{(n)} \cdot \frac{1}{\gamma} = \frac{\overline{h}^{(n)}}{1-\gamma}.\end{aligned}$$

More precisely, summing the telescoping series: $V^*(s) - v^{(n)}(s) = \lim_{N\to\infty}[v^{(N)}(s) - v^{(n)}(s)] = \sum_{k=1}^{\infty} h^{(n+k)}(s)$. Each $h^{(n+k)}(s) \leq \overline{h}^{(n+k)} \leq \gamma^k \overline{h}^{(n)}$, so $V^*(s) - v^{(n)}(s) \leq \sum_{k=1}^{\infty} \gamma^k \overline{h}^{(n)} = \frac{\gamma}{1-\gamma}\overline{h}^{(n)}$.

A similar argument using $h^{(n+k)}(s) \geq \underline{h}^{(n+k)} \geq \gamma^k \underline{h}^{(n)}$ gives $V^*(s) - v^{(n)}(s) \geq \frac{\gamma}{1-\gamma}\underline{h}^{(n)}$.

Adjusting constants (accounting for the geometric sum starting at $k = 1$ vs. $k = 0$), we obtain the bounds stated.

Lower bound. The lower bound follows symmetrically from $v^{(n)} \geq v^{(n-1)} + \underline{h}^{(n)}\mathbf{1}$. □

Improved Stopping Criterion via Span

Value iteration can be stopped when

$$\frac{\mathrm{sp}(v^{(n)} - v^{(n-1)})}{1-\gamma} < 2\varepsilon,$$

which guarantees $\|v^{(n)} - V^*\|_\infty < \varepsilon$. Compare this with the standard stopping criterion from Lesson 42: $\|v^{(n)} - v^{(n-1)}\|_\infty < \varepsilon(1-\gamma)/\gamma$.
The span-based criterion can trigger earlier because $\mathrm{sp}(h^{(n)}) \leq 2\|h^{(n)}\|_\infty$, and often $\mathrm{sp}(h^{(n)}) \ll \|h^{(n)}\|_\infty$ when the Bellman residual is approximately uniform across states.

Example 4.2 (Span vs. Sup-Norm Stopping)**.** Consider a two-state MDP with $\gamma = 0.95$, $R(s_1, a) = 10$, $R(s_2, a) = 10$ for all actions, and $P(s_1 \mid s_1, a) = 0.5$, $P(s_1 \mid s_2, a) = 0.5$ for the single action a. The value function is $V^*(s_1) = V^*(s_2) = 10/(1-0.95) = 200$ (the MDP is symmetric).

Starting from $v^{(0)} = (0,0)^\top$, after n iterations: $v^{(n)}(s_1) = v^{(n)}(s_2) = 10(1 + 0.95 + \cdots + 0.95^{n-1}) = 10 \cdot \frac{1-0.95^n}{0.05}$.

The Bellman residual is $h^{(n)}(s) = 10 \cdot 0.95^{n-1}$ for both states. Thus $\|h^{(n)}\|_\infty = 10 \cdot 0.95^{n-1}$ but $\mathrm{sp}(h^{(n)}) = 0$ for all n.

The sup-norm criterion requires $10 \cdot 0.95^{n-1} < \varepsilon \cdot 0.05/0.95$, i.e., $n > 1 + \frac{\log(10 \cdot 0.95/(\varepsilon \cdot 0.05))}{\log(1/0.95)}$. For $\varepsilon = 1$: $n > 1 + \log(190)/\log(1/0.95) \approx 103$ iterations.

The span criterion detects convergence after just 1 iteration! The span correctly recognizes that the iterates are converging uniformly, with no relative error between states.

5 Monotonicity of Value Iteration

The behavior of the VI sequence depends on the initialization $v^{(0)}$. Certain choices guarantee monotone convergence to V^*.

Theorem 5.1 (Monotone Convergence from Below)**.** *If $v^{(0)} \leq \mathcal{T}^* v^{(0)}$ componentwise (i.e., $v^{(0)}(s) \leq (\mathcal{T}^* v^{(0)})(s)$ for all s), then the value iteration sequence is monotonically non-decreasing: $v^{(0)} \leq v^{(1)} \leq v^{(2)} \leq \cdots \leq V^*$, and $v^{(n)} \uparrow V^*$.*

Proof. By hypothesis, $v^{(0)} \leq v^{(1)} = \mathcal{T}^* v^{(0)}$. By monotonicity of $\mathcal{T}^*$ (Lesson 42, Proposition 8.1): $v^{(1)} = \mathcal{T}^* v^{(0)} \leq \mathcal{T}^* v^{(1)} = v^{(2)}$. By induction, $v^{(n)} \leq v^{(n+1)}$ for all $n \geq 0$.

Since $v^{(0)} \leq V^*$ (we verify this below for the standard choice $v^{(0)} = 0$), monotonicity of $\mathcal{T}^*$ gives $v^{(n)} = (\mathcal{T}^*)^n v^{(0)} \leq (\mathcal{T}^*)^n V^* = V^*$ for all n. Combined with geometric convergence $v^{(n)} \to V^*$, we get $v^{(n)} \uparrow V^*$. □

Corollary 5.2 (Standard Initialization $v^{(0)} = 0$)**.** *If all rewards are non-negative ($R(s,a) \geq 0$ for all s, a) and $v^{(0)} = 0$, then $v^{(0)} \leq \mathcal{T}^* v^{(0)}$ and the VI sequence is monotonically non-decreasing to V^*.*

Proof. $(\mathcal{T}^* v^{(0)})(s) = \max_a [R(s,a) + \gamma \cdot 0] = \max_a R(s,a) \geq 0 = v^{(0)}(s)$. □

Theorem 5.3 (Monotone Convergence from Above)**.** *If $v^{(0)} \geq \mathcal{T}^* v^{(0)}$ componentwise, then the value iteration sequence is monotonically non-increasing: $v^{(0)} \geq v^{(1)} \geq v^{(2)} \geq \cdots \geq V^*$, and $v^{(n)} \downarrow V^*$.*

Proof. The proof mirrors that of Theorem 5.1 with all inequalities reversed. □

Corollary 5.4 (Upper-Bound Initialization)**.** *If $R(s,a) \geq 0$ for all (s,a) and $v^{(0)} = \frac{R_{\max}}{1-\gamma}\mathbf{1}$, then $v^{(0)} \geq \mathcal{T}^* v^{(0)}$ and the VI sequence is monotonically non-increasing to V^*.*

Proof. For any s:

$$(\mathcal{T}^* v^{(0)})(s) = \max_a \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \frac{R_{\max}}{1-\gamma} \right]$$
$$= \max_a R(s,a) + \frac{\gamma R_{\max}}{1-\gamma} \le R_{\max} + \frac{\gamma R_{\max}}{1-\gamma} = \frac{R_{\max}}{1-\gamma} = v^{(0)}(s). \qquad \square$$

Bracketing V^* with Monotone Sequences

Running value iteration twice – once from below with $v_L^{(0)} = 0$ (assuming $R \ge 0$) and once from above with $v_U^{(0)} = \frac{R_{\max}}{1-\gamma}\mathbf{1}$ – produces sequences satisfying

$$v_L^{(n)}(s) \le V^*(s) \le v_U^{(n)}(s) \quad \text{for all } s, n.$$

The gap $v_U^{(n)}(s) - v_L^{(n)}(s) \le \gamma^n \frac{R_{\max}}{1-\gamma}$ provides a guaranteed interval estimate. Since both sequences are monotone, partial ordering information is preserved at each step.

Example 5.5 (Monotone VI on a Three-State MDP)**.** Consider an MDP with $\mathcal{S} = \{1,2,3\}$, $\mathcal{A} = \{a\}$ (single action), $\gamma = 0.9$, $R(s,a) = s$ for all s, and $P(s' \mid s,a) = 1/3$ for all s, s' (uniform transitions). Then $V^*(s) = s + \frac{0.9}{3}[V^*(1) + V^*(2) + V^*(3)]$. By symmetry of the transition kernel, $\bar{V} = \frac{1}{3}[V^*(1) + V^*(2) + V^*(3)]$ satisfies $\bar{V} = 2 + 0.9\bar{V}$, giving $\bar{V} = 20$. Thus $V^*(s) = s + 0.9 \cdot 20/(3/3) = s + 18$. So $V^* = (19, 20, 21)^\top$.

From below ($v^{(0)} = 0$): $v^{(1)} = (1,2,3)^\top$. $v^{(2)}(s) = s + 0.9 \cdot (6/3) = s + 1.8$, so $v^{(2)} = (2.8, 3.8, 4.8)^\top$. Each iterate increases, approaching $V^* = (19, 20, 21)^\top$.

From above ($v^{(0)} = 30 \cdot \mathbf{1}$, using $R_{\max}/(1-\gamma) = 3/0.1 = 30$): $v^{(1)}(s) = s + 0.9 \cdot 30 = s + 27$, so $v^{(1)} = (28, 29, 30)^\top$. Each iterate decreases toward V^*.

The gap after $n = 1$: $v_U^{(1)} - v_L^{(1)} = (27, 27, 27)^\top$, compared to $\gamma^1 \cdot 30 = 27$. The bound is tight.

6 Policy Convergence

In practice, the greedy policy with respect to $v^{(n)}$ often stabilizes long before $v^{(n)}$ converges to V^* in norm. We formalize this observation.

Definition 6.1 (Greedy Policy Sequence)**.** The *greedy policy sequence* generated by value iteration is $\{d^{(n)}\}_{n \ge 0}$, where

$$d^{(n)}(s) = \arg\max_{a \in \mathcal{A}} \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, v^{(n)}(s') \right].$$

Theorem 6.2 (Eventual Policy Stability)**.** *There exists $N < \infty$ such that $d^{(n)} = \pi^*$ for all $n \ge N$.*

Proof. Since $v^{(n)} \to V^*$ and both $\mathcal{S}$ and $\mathcal{A}$ are finite, we use the following argument. For each state s, define the *action gap*:

$$\Delta(s) = Q^*(s, \pi^*(s)) - \max_{a \ne \pi^*(s)} Q^*(s,a).$$

If $\pi^*(s)$ is the unique optimal action at s, then $\Delta(s) > 0$. (If multiple actions are optimal, any greedy selection is acceptable.)

For any s where $\pi^*(s)$ is uniquely optimal, $|Q^{(n)}(s,a) - Q^*(s,a)| \le \gamma\|v^{(n)} - V^*\|_\infty$ for all a, where $Q^{(n)}(s,a) = R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)v^{(n)}(s')$. Once $\gamma\|v^{(n)} - V^*\|_\infty < \Delta(s)/2$, the greedy action at s with respect to $v^{(n)}$ must equal $\pi^*(s)$.

Since $\|v^{(n)} - V^*\|_\infty \to 0$, there exists N_s such that this holds for all $n \ge N_s$. Taking $N = \max_{s\in\mathcal{S}} N_s$ gives $d^{(n)} = \pi^*$ for all $n \ge N$. □

Remark 6.3. The *action gap* $\Delta(s)$ controls how many iterations are needed for the policy to stabilize at state s. States with a large gap (where the optimal action is much better than alternatives) see early policy convergence; states with a small gap (nearly tied actions) require more iterations.

Policy May Oscillate Before Convergence

Before reaching the eventual stability index N, the greedy policy $d^{(n)}$ can oscillate between different actions, especially at states where the action gap is small. A policy-based stopping criterion (stop when $d^{(n)} = d^{(n-1)}$ for several consecutive iterations) is heuristic and not guaranteed to detect optimality without additional checks. The value-based criterion remains the only provably correct stopping rule.

Proposition 6.4 (Policy Convergence Rate)**.** *Let $\Delta_{\min} = \min_{s:\Delta(s)>0} \Delta(s)$ be the minimum positive action gap. The greedy policy converges to π^* after at most*

$$N = \left\lceil \frac{1}{\log(1/\gamma)} \log\left(\frac{2\gamma R_{\max}}{\Delta_{\min}(1-\gamma)}\right)\right\rceil$$

iterations (starting from $v^{(0)} = 0$).

Proof. We need $\gamma\|v^{(n)} - V^*\|_\infty < \Delta_{\min}/2$. Using the a priori bound $\|v^{(n)} - V^*\|_\infty \le \gamma^n R_{\max}/(1-\gamma)$, it suffices to have $\gamma^{n+1}R_{\max}/(1-\gamma) < \Delta_{\min}/2$. Solving for n gives the stated bound. □

Example 6.5 (Early Policy Convergence)**.** For the two-state MDP from Lesson 42 (Example 5.5) with $\gamma = 0.5$, the optimal actions are $\pi^*(s_1) = a_2$, $\pi^*(s_2) = a_2$. After just one VI iteration ($v^{(1)} = (10, 2)^\top$), the greedy policy is already $d^{(1)}(s_1) = a_2$ (since $10 > 5 + 0.5(0.5\cdot 10 + 0.5\cdot 2) = 8$, verifying via $Q^{(1)}$) and $d^{(1)}(s_2) = a_2$ (since $4.2 > 3.2$). The policy converges after 1 iteration, while the value function requires many more iterations for high accuracy.

7 Refined Error Bounds via Bellman Residuals

The Bellman residual $\mathcal{T}^*v - v$ at any iterate provides information about the distance to V^*.

Theorem 7.1 (Bellman Residual Bound)**.** *For any $v \in \mathbb{R}^S$, let $r_v = \mathcal{T}^*v - v$ be the Bellman residual. Then*

$$\frac{\min_s r_v(s)}{1-\gamma} \le V^*(s) - v(s) \le \frac{\max_s r_v(s)}{1-\gamma} \quad \textit{for all } s \in \mathcal{S}.$$

Proof. We prove the upper bound; the lower bound is analogous.

Since $v + \max_s r_v(s)\cdot\mathbf{1} \ge v + r_v = \mathcal{T}^*v$, i.e., $v + (\max_s r_v(s))\mathbf{1} \ge \mathcal{T}^*v$, applying $\mathcal{T}^*$ and using monotonicity and the shift property:

$$\begin{aligned}
\mathcal{T}^*(v + (\max_s r_v(s))\mathbf{1}) &= \mathcal{T}^*v + \gamma \max_s r_v(s)\cdot\mathbf{1} \\
&= v + r_v + \gamma \max_s r_v(s)\cdot\mathbf{1} \\
&\le v + \max_s r_v(s)\cdot\mathbf{1} + \gamma \max_s r_v(s)\cdot\mathbf{1}.
\end{aligned}$$

Wait – we need a cleaner argument. Define $\bar{v} = v + \frac{\max_s r_v(s)}{1-\gamma}\mathbf{1}$. We claim $\bar{v} \geq \mathcal{T}^*\bar{v}$. Indeed:

$$\begin{aligned}(\mathcal{T}^*\bar{v})(s) &= (\mathcal{T}^*v)(s) + \frac{\gamma \max_s r_v(s)}{1-\gamma} \\ &= v(s) + r_v(s) + \frac{\gamma \max_s r_v(s)}{1-\gamma} \\ &\leq v(s) + \max_s r_v(s) + \frac{\gamma \max_s r_v(s)}{1-\gamma} \\ &= v(s) + \frac{\max_s r_v(s)}{1-\gamma} = \bar{v}(s).\end{aligned}$$

Since $\bar{v} \geq \mathcal{T}^*\bar{v}$, by Theorem 5.3 (monotone convergence from above), iterating $\mathcal{T}^*$ from $\bar{v}$ gives a non-increasing sequence converging to V^*. Thus $V^* \leq \bar{v}$, i.e., $V^*(s) \leq v(s) + \frac{\max_s r_v(s)}{1-\gamma}$.

The lower bound follows by a symmetric argument using $\min_s r_v(s)$. □

Corollary 7.2 (Two-Sided Bound from Bellman Residual)**.** *For any $v \in \mathbb{R}^S$:*

$$\|v - V^*\|_\infty \leq \frac{\|\mathcal{T}^*v - v\|_\infty}{1-\gamma}.$$

Proof. From Theorem 7.1, $|V^*(s) - v(s)| \leq \max\{|\max_s r_v(s)|, |\min_s r_v(s)|\}/(1-\gamma) \leq \|r_v\|_\infty/(1-\gamma)$. □

Bellman Error in Approximate DP

In approximate dynamic programming and deep RL, the Bellman residual $\|\mathcal{T}^*\hat{v} - \hat{v}\|_\infty$ (or its expected version $\mathbb{E}[(\mathcal{T}^*\hat{v}(S) - \hat{v}(S))^2]$) is used as a loss function for training value function approximators. The bound in Corollary 7.2 shows that minimizing the Bellman residual controls the distance to V^*. This is the theoretical basis for the mean-squared Bellman error (MSBE) objective used in TD learning and fitted value iteration.

8 Exercises

Exercise 8.1 (Span Computation)**.** $(*)$
For the vectors $u = (3, 7, 5, 1)^\top$ and $v = (2, 4, 6, 0)^\top$, compute:

(a) $\mathrm{sp}(u)$, $\mathrm{sp}(v)$, $\mathrm{sp}(u - v)$.

(b) $\|u - v\|_\infty$ and verify that $\mathrm{sp}(u - v) \leq 2\|u - v\|_\infty$.

(c) $\mathrm{sp}(u + 10 \cdot \mathbf{1})$ and verify translation invariance.

Exercise 8.2 (Monotone VI Verification)**.** $(*)$
For the three-state MDP in Example 5.5, compute $v_L^{(3)}$ (from below) and $v_U^{(3)}$ (from above) and verify that $v_L^{(3)} \leq V^* \leq v_U^{(3)}$. Compute the gap $v_U^{(3)} - v_L^{(3)}$ and compare with $\gamma^3 \cdot 30 \cdot \mathbf{1}$.

Exercise 8.3 (Span-Based Stopping in Practice)**.** $(**)$
Consider an MDP with $\gamma = 0.95$ and $R_{\max} = 1$. Suppose after n iterations of VI, $v^{(n)} - v^{(n-1)} = (0.1, 0.1, 0.1, 0.12, 0.09)^\top$.

(a) Compute the standard stopping bound $\frac{\gamma}{1-\gamma}\|v^{(n)} - v^{(n-1)}\|_\infty$.

(b) Compute the span-based bound $\frac{\mathrm{sp}(v^{(n)} - v^{(n-1)})}{2(1-\gamma)}$.

(c) By what factor is the span-based bound tighter?

Exercise 8.4 (Bellman Residual Bounds)**.** (∗∗)
Let $v = (10, 8, 5)^\top$ be an approximate value function for a three-state MDP with $\gamma = 0.9$. Suppose the Bellman backup gives $\mathcal{T}^* v = (10.5, 8.2, 5.8)^\top$.

(a) Compute the Bellman residual $r_v = \mathcal{T}^* v - v$.

(b) Use Theorem 7.1 to bound $V^*(s) - v(s)$ for each s.

(c) Use Corollary 7.2 to bound $\|v - V^*\|_\infty$.

(d) If the greedy policy has action gap $\Delta_{\min} = 1$, is v accurate enough to guarantee the correct greedy policy?

Exercise 8.5 (Span Contraction for $\mathcal{T}^\pi$)**.** (∗ ∗ ∗)
Prove that the Bellman expectation operator $\mathcal{T}^\pi$ satisfies $\operatorname{sp}(\mathcal{T}^\pi u - \mathcal{T}^\pi v) \le \gamma \operatorname{sp}(u - v)$ for any stationary policy π.

Hint: Adapt the proof of Theorem 3.4, replacing the max over actions with the weighted sum under π.

Exercise 8.6 (Action Gap and Approximation)**.** (∗ ∗ ∗)
Consider a family of MDPs parameterized by $\delta > 0$ where the action gap at some state s is $\Delta(s) = \delta$.

(a) Show that the number of VI iterations needed for the greedy policy to stabilize at state s is $\Theta(\frac{1}{1-\gamma} \log(R_{\max}/(\delta(1-\gamma))))$.

(b) Construct a two-state, two-action MDP where $\Delta(s) = \delta$ for a given $\delta > 0$ and verify the bound from (a).

(c) Discuss the implications for RL: what happens in environments where many states have near-zero action gaps?

9 Summary

- The *span seminorm* $\operatorname{sp}(v) = \max_s v(s) - \min_s v(s)$ is translation invariant and satisfies $\operatorname{sp}(\mathcal{T}^* u - \mathcal{T}^* v) \le \gamma \operatorname{sp}(u - v)$.
- The span-based stopping criterion $\operatorname{sp}(v^{(n)} - v^{(n-1)})/(1-\gamma) < 2\varepsilon$ can be significantly tighter than the standard $\|v^{(n)} - v^{(n-1)}\|_\infty$ criterion, especially when the Bellman residual is approximately uniform.
- *Monotone convergence*: VI from below ($v^{(0)} \le \mathcal{T}^* v^{(0)}$) gives a non-decreasing sequence; VI from above ($v^{(0)} \ge \mathcal{T}^* v^{(0)}$) gives a non-increasing sequence. Running both brackets V^*.
- The *greedy policy* converges to π^* in finitely many iterations; the rate depends on the minimum *action gap* $\Delta_{\min}$ and can be much faster than value convergence.
- The *Bellman residual* $r_v = \mathcal{T}^* v - v$ provides two-sided bounds: $\frac{\min_s r_v(s)}{1-\gamma} \le V^*(s) - v(s) \le \frac{\max_s r_v(s)}{1-\gamma}$, yielding $\|v - V^*\|_\infty \le \|r_v\|_\infty/(1-\gamma)$.
- Policy convergence can occur much earlier than value convergence, but detecting policy convergence reliably requires value-based verification.

10 Further Reading

- **Lesson 44** extends the MDP framework to the average reward criterion, where the span seminorm plays an even more central role (the Bellman equation for average reward is defined only up to a constant shift).
- **Lesson on TD Learning** shows how value iteration ideas translate to the model-free setting, where Bellman backups use sampled transitions rather than expected values.
- Puterman [2014, Chapter 6, Section 6.6] develops acceleration techniques for value iteration, including MacQueen's bounds and Gauss-Seidel (asynchronous) updates.
- Bertsekas [2012, Chapter 2] extends error bound analysis to approximate dynamic programming with function approximation, deriving performance bounds for fitted value iteration.
- Denardo [1967] provides the abstract framework for contraction operators on ordered spaces, generalizing the span contraction results to broader classes of dynamic programming problems.

References

Dimitri P. Bertsekas. *Dynamic Programming and Optimal Control*, volume II. Athena Scientific, 4th edition, 2012.

Eric V. Denardo. Contraction mappings in the theory underlying dynamic programming. *SIAM Review*, 9(2):165–177, 1967.

Martin L. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. John Wiley & Sons, 2nd edition, 2014.

Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2nd edition, 2018.